James Mann

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Education

University College London | **Computer Science MEng** | **Sept 2024 – May 2027**

- First year average: 81%
- First Year Representative of the AI Society

Experience

MATS | **Neel Nanda's MATS Training Phase** | **Sep 2025 - Oct 2025**

- Worked on various mini-projects including an LLM progress bar based on model internals and a project investigating LLM's latent representations of user characteristics (age, sex, star sign).
- Developed a final project investigating model tendencies to reward hack, identifying that a deception direction could control gpt-oss-20b's reward hacking.
- Trained linear probes to catch reward hacking intent. Reconstructed the found directions with an SAE to surface hypotheses on causes.

Arcadia Impact | **Research Analyst / Software Engineer** | **May 2025 -**

- Led the creation of a technical tender responding to a €1,000,000 tender for AI Evaluations Infrastructure for the EU, ultimately coming second out of fourteen applying consortiums.
- Part of the maintenance team for Inspect Evals, responding to issues and applying QA to prospective new Evaluations.
- Independently developed cost modeling software that simulated 10,000+ instances of the company, accounting for underlying randomness, leading to 15% better resource allocation.

Bluedot Impact | **Incubator Week + Hackathon Prize** | **Nov 2025**

- Accepted into v1 of Bluedot's Incubator Week
- Spent a week developing a plan to create tailored probes for inference-time monitoring.
- Came 2nd overall, winnning £1000 in Bluedot's 100+ person hackathon solo developing an MVP, creating CUDA kernels to minimise latency of monitoring, eventually optimising it down to <5%.

Impact Research Groups | **Technical AI Safety Research Sprint** | **Feb 2025 - April 2025**

- First author of a paper produced as part of an 8 week competitive research sprint, coming third.
- Identified a safety-critical latent factor in high-dimensional model activation space that could steer between refusal and compliance.
- Proposed a theoretical framework for the success of jailbreaks as suppressors of harmfulness perception.

Projects

Non-deterministic sampling investigation

- Utilised multivariate regression to identify and controlling planning capabilities in model latent space.
- Linear probes trained on input activations to predict the non-deterministic, sampling-induced distribution, utilizing the Kullback-Leibler divergence function, achieving a 0.95 R^2.
- Applied in-place, input activation steering to control the high-level length feature i.e. make the model output longer responses, or make the model output shorter responses, without affecting coherence.

Probabilistic Image Generation

- Implemented a denoising diffusion probabilistic model in Pytorch, trained on CIFAR100 to generate novel images from the distribution, iteratively optimising the hyperparameters to improve the plausibility of samples.
- Followed the paper Denoising Diffusion Probabilistic Models, implementing a UNet with attention heads